Chapter 18

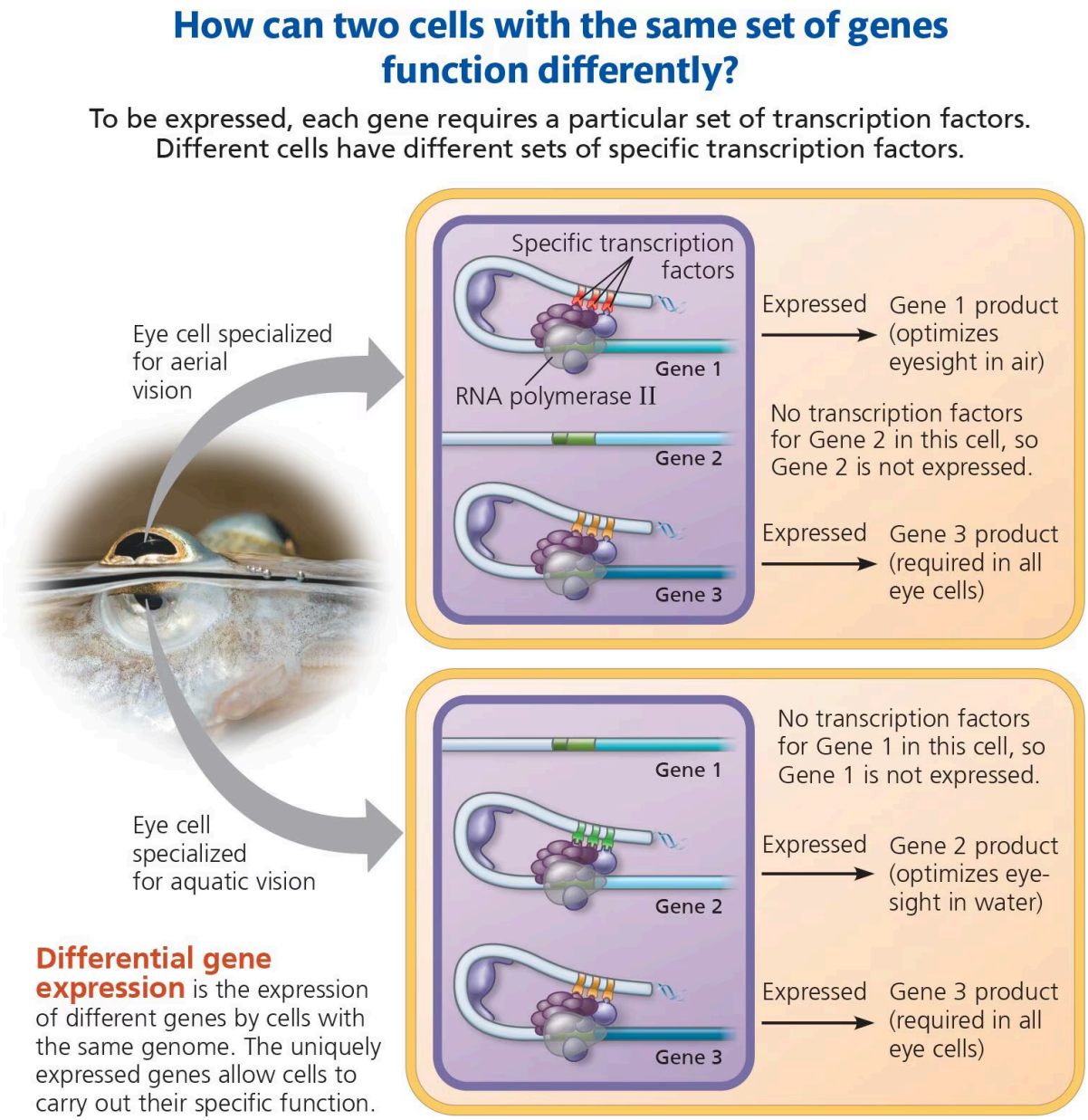
Campbell Biology, AP Edition, 12e

Chapter 18

Regulation of Gene Expression

Figure 18.1

Anableps anableps, or “cuatro ojos” (“four eyes”), glides through lakes and ponds in South America, the upper half of each eye protruding from the water. The eye’s upper half is well-suited for aerial vision and the lower half for aquatic vision. All eye cells, however, contain the same genes.



To be expressed, each gene requires a particular set of transcription factors. Different cells have different sets of specific transcription factors. An Anableps has eye cells specialized for aerial vision and eye cells specialized for aquatic vision. Each cell has genes 1, 2, and 3. For the aerial cells, genes 1 and 3 are expressed where gene 1 produces a product for optimizing eyesight in air, and gene 3 is required in all eye cells. Gene 1 has specific transcription factors, and no transcription factors are present for gene 2, so gene 2 is not expressed. For the aquatic cells, gene 1 has no specific transcription factors, so it isn't expressed, but gene 2 does, so it is expressed. Gene 3 is also expressed.

Key Concepts

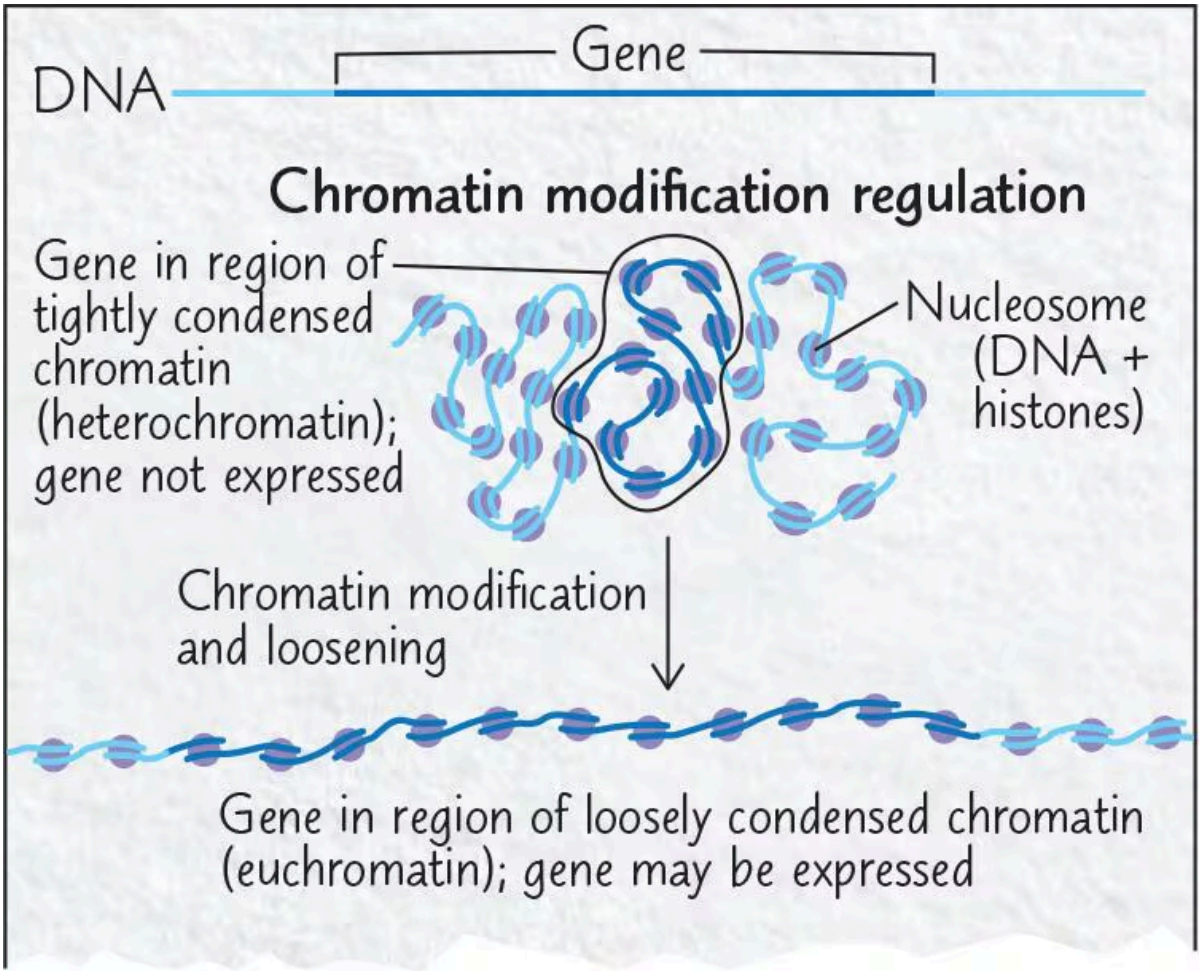
- 18.1**Bacteria often respond to environmental change by regulating transcription**

- 18.2**Eukaryotic gene expression is regulated at many stages**
- 18.3**Noncoding RNAs play multiple roles in controlling gene expression**
- 18.4**A program of differential gene expression leads to the different cell types in a multicellular organism**
- 18.5**Cancer results from genetic changes that affect cell cycle control**

Organisms use feedback mechanisms and transcription factors to maintain dynamic homeostasis through differential gene expression **(BIG IDEAS 2 & 3)**, conserved core processes, and features that are shared by all organisms **(BIG IDEA 1)**.

Study Tip

Make a visual study guide: Draw a region of DNA representing a gene. For each level of gene regulation that you read about, draw a sketch of your gene being regulated at that level. A drawing of chromatin modification regulation is shown as an example.



On the D N A is a gene for chromatin modification regulation. The gene is in the region of tightly condensed chromatin (heterochromatin gene); gene not expressed. Nucleosome (D N A plus histones) surrounds this region. The chromatin modification and loosening occurs. The gene is in a region of loosely condensed chromatin (euchromatin); gene may be expressed.

Get Ready for This Chapter

In this chapter, you will learn how bacterial and eukaryotic genes are regulated, including the role of noncoding RNAs. The role of gene regulation in embryonic development is then described, followed by a discussion of cancer as an example of gene regulation gone awry. Before beginning this chapter, you should be able to do the following:

- Describe the difference between euchromatin and heterochromatin (see Concept 16.3 and Figure 16.23).
- Describe the process of transcription (see Concept 17.2 and Figure 17.8).
- Describe the process of translation (see Concept 17.4, Figure 17.19, Figure 17.20, and Figure 17.21).

- Explain cell signaling (see Concept 11.1, Concept 11.4, and Figure 11.6).

The following questions provide a quick check of your basic knowledge in these areas.

1. How is euchromatin different from heterochromatin?

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Heterochromatin is the more densely arranged chromatin that is found during mi
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That's incorrect. Try again.

Incorrect. Review the following material: The 10-nm fiber is DNA wound around histones in nucleosomes, connected by linker DNA. During interphase, the 10-nm fiber is found in two arrangements: loosely arranged in euchromatin, where genes are more accessible to enzymes that transcribe genes, and densely arranged in heterochromatin, where genes are typically inaccessible to transcription enzymes.

Both consist of the 10-nm fiber, which during interphase is more densely arranged in euchromatin, and more loosely arranged in heterochromatin.

That's incorrect. Try again.

Incorrect. Review the following material: The 10-nm fiber is DNA wound around histones in nucleosomes, connected by linker DNA. During interphase, the 10-nm fiber is found in two arrangements: loosely arranged in euchromatin, where genes are more accessible to enzymes that transcribe genes, and densely arranged in heterochromatin, where genes are typically inaccessible to transcription enzymes.

Both consist of the 10-nm fiber, which during interphase is more densely arranged in heterochromatin, and more loosely arranged in euchromatin.

That's correct.

Correct! If you had trouble with this question, review the following material: The 10-nm fiber is DNA wound around histones in nucleosomes, connected by linker DNA. During interphase, the 10-nm fiber is found in two arrangements: loosely arranged in euchromatin, where genes are more accessible to enzymes that transcribe genes, and densely arranged in heterochromatin, where genes are typically inaccessible to transcription enzymes.

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Heterochromatin is accessible to the enzymes responsible for transcribing genes, wh
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That's incorrect. Try again.

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arranged in heterochromatin, where genes are typically inaccessible to transcription enzymes.

2. Which of the following are required for the process of transcription?

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transcription factors and amino acids
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That's incorrect. Try again.

Incorrect. Review the following material: In the process of transcription, RNA polymerase binds in a precise location and orientation on the promoter, initiating RNA synthesis at the start point on the DNA template strand. In eukaryotes, a collection of proteins called transcription factors mediate the binding of RNA polymerase and the initiation of transcription. Translation is the process where tRNA is key to translating a given mRNA codon into a specific amino acid.

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tRNA and RNA polymerase
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That's incorrect. Try again.

Incorrect. Review the following material: In the process of transcription, RNA polymerase binds in a precise location and orientation on the promoter, initiating RNA synthesis at the start point on the DNA template strand. In eukaryotes, a collection of proteins called transcription factors mediate the binding of RNA polymerase and the initiation of transcription. Translation is the process where tRNA is key to translating a given mRNA codon into a specific amino acid.

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DNA and DNA polymerase
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That's incorrect. Try again.

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RNA polymerase and the promoter
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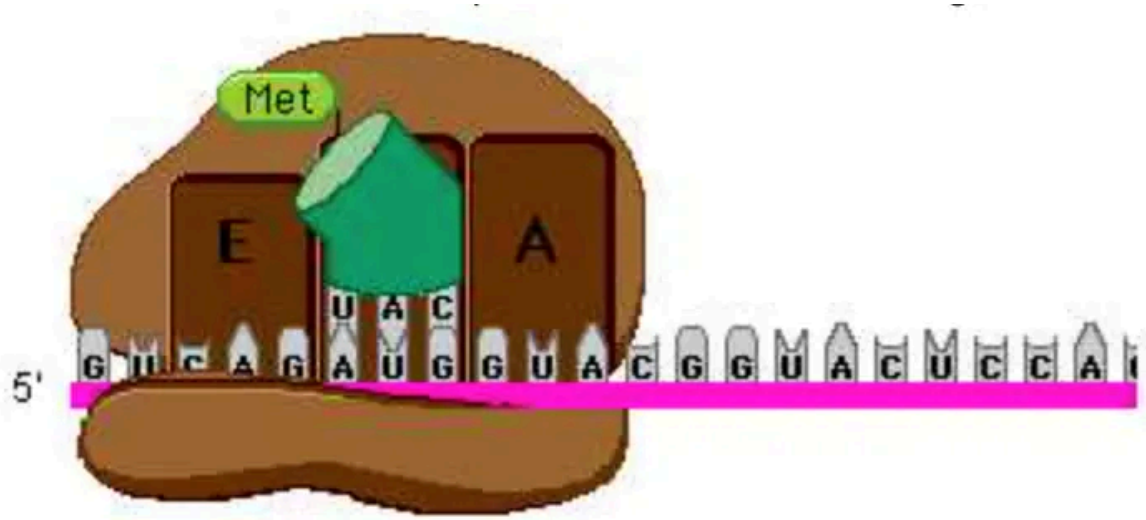
That's correct.

Correct!If you had trouble with this question, review the following material: In the process of transcription, RNA polymerase binds in a precise location and orientation on the promoter, initiating RNA synthesis at the start point on the DNA template strand. In eukaryotes, a collection of proteins called transcription factors mediate the binding of RNA polymerase and the initiation of transcription. Translation is the process where tRNA is key to translating a given mRNA codon into a specific amino acid.



3. What process is shown in the diagram?

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initiation of translation
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That's correct.

Correct! If you had trouble with this question, review the following material: The initiation stage of translation brings together mRNA, a tRNA bearing the first amino acid of the polypeptide, and the two subunits of a ribosome. First, a small ribosomal subunit binds to both mRNA and a specific initiator tRNA, which carries the amino acid methionine. In the elongation stage of translation, amino acids are added one by one to the previous amino acid at the C-terminus of the growing polypeptide chain. Codon recognition occurs when the anticodon of an incoming aminoacyl tRNA base-pairs with the complementary mRNA codon in the A site of the ribosome. In the peptide bond formation step of elongation, an rRNA molecule of the large ribosomal subunit catalyzes the formation of a peptide bond between the amino group of the new amino acid in the A site and the carboxyl end of the growing polypeptide in the P site.

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codon recognition
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That's incorrect. Try again.

Incorrect. Review the following material: The initiation stage of translation brings together mRNA, a tRNA bearing the first amino acid of the polypeptide, and the two subunits of a ribosome. First, a small ribosomal subunit binds to both mRNA and a specific initiator tRNA, which carries the amino acid methionine. In the elongation stage of translation, amino acids are added one by one to the previous amino acid at the C-terminus of the growing polypeptide chain. Codon recognition occurs when the anticodon of an incoming aminoacyl tRNA base-pairs with the complementary mRNA codon in the A site of the ribosome. In the peptide bond formation step of elongation, an rRNA molecule of the large ribosomal subunit catalyzes the formation of a peptide bond between the amino group of the new amino acid in the A site and the carboxyl end of the growing polypeptide in the P site.

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peptide bond formation
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That's incorrect. Try again.

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elongation
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That's incorrect. Try again.

Incorrect. Review the following material: The initiation stage of translation brings together mRNA, a tRNA bearing the first amino acid of the polypeptide, and the two subunits of a ribosome. First, a small ribosomal subunit binds to both mRNA and a specific initiator tRNA, which carries the amino acid methionine. In the elongation stage of translation, amino acids are added one by one to the previous amino acid at the C-terminus of the growing polypeptide chain. Codon recognition occurs when the anticodon of an incoming aminoacyl tRNA base-pairs with the complementary mRNA codon in the A site of the ribosome. In the peptide bond formation step of elongation, an rRNA molecule of the large ribosomal subunit catalyzes the formation of a peptide bond between the amino group of the new amino acid in the A site and the carboxyl end of the growing polypeptide in the P site.

4. When a neuron responds to a particular neurotransmitter molecule by opening gated ion channels, the neurotransmitter molecules are serving as which part of the signal pathway?

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response molecule
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That's incorrect. Try again.

Incorrect. Review the following material: Neurotransmitter molecules act as chemical signals, diffusing across the synapse and triggering a response in the target cell. The ability of a cell to respond is determined by whether it has a specific receptor molecule that can bind to the signaling molecule. The information conveyed by the binding of a specific signaling molecule to a receptor must then be changed into another form (transduced) inside the cell before the cell can respond.

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signaling receptor
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That's incorrect. Try again.

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transducing, or relay, molecule
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That's incorrect. Try again.

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signaling molecule
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That's correct.

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molecule. The information conveyed by the binding of a specific signaling molecule to a receptor must then be changed into another form (transduced) inside the cell before the cell can respond.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 18.1Explain how the trp and lac operons function.
- 18.2Describe the stages of eukaryotic gene expression and the regulation that can occur at each stage.
- 18.3Define "noncoding RNAs" and explain how they participate in regulating gene expression, including their effects on chromatin.
- 18.4Explain how differential gene expression leads to the different cell types in a multicellular organism.
- 18.5Describe how cancer can result from genetic changes that affect cell cycle control.